Supplementary results and verification records

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This supplement accompanies the new study of three independently fine-tuned starting pairs. It contains every fixed endpoint, including unfavorable results. It does not pool earlier single-starting-pair experiments. Epoch 40 is primary; epoch 20 lies on the same 40-epoch schedule. All displayed AP values use a 0–100 scale, and differences use percentage points. Tables are rounded for readability; final_all_pair_analysis.json retains 960 unrounded metric values.

S1 Reading the endpoint tables

Column	Definition
BDD native	Ten classes; unchanged Ultralytics AP50:95.
BDD common	Person/car; shared COCO AP101 at IoUs 0.50–0.95.
KITTI KI	Shared two-class AP101 with source-ignore projection.
KITTI KN	Same common metric without ignore records; sensitivity only.
KITTI native	Mean Car/Pedestrian moderate native AP_R40.

All-class inference precedes mapping. Shared classes and integration do not eliminate annotation, ignore or scene differences. KITTI is the previously examined 7,481-image public labeled training partition. BDD validation is reused and based on the reconstructed legacy mirror. Neither dataset is described as a fresh untouched test.

S2 Evidence chain

A frozen roster binds six base and 42 continuation checkpoints. Each completed endpoint binds all six execution steps; each numerical proof binds its completion marker, score artifacts and audit source. The completion manifests cover all 48 IDs exactly. Final numerical replay finished on 21 October 2026 at 00:05:48 UTC. A separate local arithmetic implementation passed at 00:12:40 UTC without importing the aggregation implementation or loading a model.

Protocol: c2de04c650b0a3553aef88712a8cc4b8521fe8dbdd4f8fee589766248da1c8ba

Roster: 4a5afdd71ff4ac306e77110019a222ff8b42b184a38d8aa08f3464a59a5e0a53

Plan: 9eb59cda78a1512a732b068cf172dd74009eb70232ddcbcceba32f6f82d5fc2d

Analysis: 6ba83713bdf8ea9b050beda6a5055dcc214e4d613927d3904e39831abd4ff5b9

Software tests verify implementations under known inputs, and hash checks verify file identity. Neither establishes external validity or independent authentication of the label source. Raw copied training records, per-image data and private transport configuration are excluded from the public package.

S3 All endpoints for pair 101

Table S1. All 16 endpoints for starting pair 101. The first two rows are the epoch-100 bases; continuation labels give objective, coefficient and epoch. Scores are AP × 100.

Endpoint	BDD native	BDD common	KITTI KI	KITTI KN	KITTI native
Base nano 100	25.8586	35.2175	39.1332	37.1283	71.5275
Base large 100	33.2348	42.7210	42.1437	39.2873	76.7151
Control e20	25.8843	35.2895	38.6792	36.6612	70.9084
Control e40	25.9352	35.3375	38.6431	36.6164	70.9391
MSE 0.03 e20	25.9006	35.2797	38.6755	36.6549	70.9553
MSE 0.03 e40	25.9552	35.3314	38.6011	36.5821	70.9264
MSE 0.3 e20	25.8396	35.2747	38.7099	36.7224	71.0842
MSE 0.3 e40	25.9003	35.2983	38.6861	36.7004	71.0520
MSE 3 e20	25.6745	35.2098	38.7178	36.7112	71.0588
MSE 3 e40	25.7024	35.2404	38.6312	36.6406	71.0081
CWD 0.03 e20	25.8835	35.2855	38.6995	36.6839	70.9225
CWD 0.03 e40	25.9432	35.3305	38.6163	36.5920	70.9335
CWD 0.3 e20	25.8760	35.2786	38.7120	36.7285	70.9959
CWD 0.3 e40	25.9012	35.3173	38.6290	36.6461	71.0226
CWD 3 e20	25.7183	35.2195	38.6682	36.6750	70.8980
CWD 3 e40	25.7665	35.2503	38.6008	36.6123	70.9233

The same starting nano checkpoint is used by all seven arms within this pair. The corresponding large checkpoint is the teacher. Repeated rows across endpoints represent distinct fixed checkpoint times; they are not extra independent training pairs.

The JSON companion additionally retains native BDD AP50, precision and recall, common per-class AP, native KITTI class/difficulty AP and every checkpoint SHA-256. No endpoint was selected by its score.

S3 All endpoints for pair 202

Table S2. All 16 endpoints for starting pair 202. The first two rows are the epoch-100 bases; continuation labels give objective, coefficient and epoch. Scores are AP × 100.

Endpoint	BDD native	BDD common	KITTI KI	KITTI KN	KITTI native
Base nano 100	26.0307	35.1359	38.9261	36.9819	71.2276
Base large 100	33.4108	42.7041	42.3368	39.5183	76.9842
Control e20	26.0518	35.2443	38.5297	36.5745	70.7883
Control e40	26.1218	35.2693	38.5435	36.5919	70.7692
MSE 0.03 e20	26.0417	35.3035	38.5513	36.5768	70.7659
MSE 0.03 e40	26.0966	35.3474	38.5547	36.5845	70.7780
MSE 0.3 e20	26.0325	35.3013	38.5479	36.5728	70.6534
MSE 0.3 e40	26.0833	35.3369	38.5069	36.5392	70.5969
MSE 3 e20	25.6746	35.1419	38.4119	36.4694	70.5291
MSE 3 e40	25.7423	35.1592	38.3754	36.4389	70.5811
CWD 0.03 e20	26.0389	35.3039	38.5522	36.5800	70.7874
CWD 0.03 e40	26.1240	35.3474	38.5269	36.5430	70.7617
CWD 0.3 e20	26.0442	35.2920	38.5177	36.5364	70.7583
CWD 0.3 e40	25.9934	35.3362	38.4645	36.4735	70.6670
CWD 3 e20	25.7011	35.2369	38.4770	36.5008	70.5769
CWD 3 e40	25.7209	35.2275	38.4019	36.4189	70.5571

The same starting nano checkpoint is used by all seven arms within this pair. The corresponding large checkpoint is the teacher. Repeated rows across endpoints represent distinct fixed checkpoint times; they are not extra independent training pairs.

The JSON companion additionally retains native BDD AP50, precision and recall, common per-class AP, native KITTI class/difficulty AP and every checkpoint SHA-256. No endpoint was selected by its score.

S3 All endpoints for pair 303

Table S3. All 16 endpoints for starting pair 303. The first two rows are the epoch-100 bases; continuation labels give objective, coefficient and epoch. Scores are AP × 100.

Endpoint	BDD native	BDD common	KITTI KI	KITTI KN	KITTI native
Base nano 100	25.7903	35.2781	38.7608	36.7364	70.7521
Base large 100	33.1528	42.7271	42.1190	39.3247	76.4957
Control e20	25.7842	35.3837	38.3429	36.3357	70.2295
Control e40	25.7895	35.4089	38.2740	36.2764	70.1601
MSE 0.03 e20	25.7767	35.3754	38.3302	36.3186	70.1869
MSE 0.03 e40	25.7940	35.4022	38.2651	36.2690	70.1901
MSE 0.3 e20	25.7508	35.3961	38.3430	36.3234	70.3584
MSE 0.3 e40	25.7754	35.4237	38.2739	36.2689	70.3454
MSE 3 e20	25.5389	35.2276	38.4133	36.3984	70.4429
MSE 3 e40	25.6227	35.2570	38.3341	36.3186	70.5154
CWD 0.03 e20	25.7751	35.3771	38.3098	36.2978	70.2044
CWD 0.03 e40	25.8083	35.3971	38.2612	36.2608	70.1293
CWD 0.3 e20	25.7754	35.3835	38.4559	36.4337	70.4045
CWD 0.3 e40	25.8028	35.4111	38.3389	36.3245	70.3834
CWD 3 e20	25.6617	35.2856	38.4994	36.4688	70.5843
CWD 3 e40	25.6836	35.2710	38.4876	36.4693	70.7063

The same starting nano checkpoint is used by all seven arms within this pair. The corresponding large checkpoint is the teacher. Repeated rows across endpoints represent distinct fixed checkpoint times; they are not extra independent training pairs.

The JSON companion additionally retains native BDD AP50, precision and recall, common per-class AP, native KITTI class/difficulty AP and every checkpoint SHA-256. No endpoint was selected by its score.

S4 Native KITTI detail for pair 101

Table S4. Native AP_R40 × 100 for every endpoint. E, M and H denote easy, moderate and hard. This uses native class-specific overlaps and difficulty/ignore behavior, not common AP101.

Endpoint	Car E	Car M	Car H	Ped E	Ped M	Ped H
Base nano 100	88.962	78.082	64.163	73.795	64.973	57.451
Base large 100	89.180	79.971	66.033	81.777	73.460	65.125
Control e20	88.800	77.673	63.735	73.047	64.144	56.664
Control e40	88.857	77.673	63.765	73.146	64.205	56.783
MSE 0.03 e20	88.860	77.727	63.822	73.063	64.183	56.647
MSE 0.03 e40	88.923	77.746	63.823	73.031	64.106	56.679
MSE 0.3 e20	88.858	77.751	63.853	73.286	64.417	56.866
MSE 0.3 e40	88.788	77.812	63.968	73.360	64.292	56.824
MSE 3 e20	88.668	77.734	63.962	73.597	64.383	56.970
MSE 3 e40	88.787	77.804	63.968	73.378	64.213	56.861
CWD 0.03 e20	88.768	77.679	63.771	73.045	64.166	56.689
CWD 0.03 e40	88.870	77.706	63.772	73.056	64.161	56.703
CWD 0.3 e20	88.741	77.697	63.771	73.169	64.295	56.797
CWD 0.3 e40	88.810	77.793	63.800	73.293	64.253	56.736
CWD 3 e20	88.752	77.750	63.959	73.005	64.046	56.639
CWD 3 e40	88.714	77.778	64.036	73.015	64.068	56.700

The primary native KITTI summary is the arithmetic mean of Car M and Ped M; all remaining difficulty scores are preserved here. Detection outputs came from the same immutable KITTI prediction archive used for the common views. The numerical audit checked exact export/label bindings and recomputed AP_R40 from saved precision curves; it did not rerun native C++ matching.

S4 Native KITTI detail for pair 202

Table S5. Native AP_R40 × 100 for every endpoint. E, M and H denote easy, moderate and hard. This uses native class-specific overlaps and difficulty/ignore behavior, not common AP101.

Endpoint	Car E	Car M	Car H	Ped E	Ped M	Ped H
Base nano 100	89.160	78.422	64.467	73.030	64.033	56.378
Base large 100	89.181	79.717	65.736	82.330	74.251	65.836
Control e20	88.497	77.784	64.043	72.958	63.792	56.259
Control e40	88.414	77.754	64.016	73.070	63.784	56.235
MSE 0.03 e20	88.588	77.841	64.147	72.909	63.691	56.161
MSE 0.03 e40	88.496	77.811	64.116	72.980	63.745	56.153
MSE 0.3 e20	88.609	77.844	64.059	72.734	63.463	56.043
MSE 0.3 e40	88.484	77.785	64.057	72.632	63.409	55.854
MSE 3 e20	88.439	77.829	64.114	72.429	63.229	55.712
MSE 3 e40	88.414	77.824	64.056	72.404	63.339	55.793
CWD 0.03 e20	88.464	77.763	64.029	73.034	63.811	56.295
CWD 0.03 e40	88.412	77.772	64.059	73.001	63.751	56.174
CWD 0.3 e20	88.525	77.761	64.063	72.948	63.756	56.194
CWD 0.3 e40	88.376	77.764	64.045	72.810	63.570	56.054
CWD 3 e20	88.396	77.792	64.151	72.716	63.362	55.938
CWD 3 e40	88.306	77.800	64.113	72.454	63.314	55.881

The primary native KITTI summary is the arithmetic mean of Car M and Ped M; all remaining difficulty scores are preserved here. Detection outputs came from the same immutable KITTI prediction archive used for the common views. The numerical audit checked exact export/label bindings and recomputed AP_R40 from saved precision curves; it did not rerun native C++ matching.

S4 Native KITTI detail for pair 303

Table S6. Native AP_R40 × 100 for every endpoint. E, M and H denote easy, moderate and hard. This uses native class-specific overlaps and difficulty/ignore behavior, not common AP101.

Endpoint	Car E	Car M	Car H	Ped E	Ped M	Ped H
Base nano 100	88.801	77.794	63.783	72.380	63.710	56.198
Base large 100	88.279	79.528	65.580	81.751	73.463	64.947
Control e20	88.110	77.362	63.414	72.161	63.097	55.686
Control e40	88.129	77.334	63.302	71.762	62.986	55.597
MSE 0.03 e20	88.207	77.351	63.353	71.957	63.023	55.590
MSE 0.03 e40	88.148	77.338	63.297	71.888	63.042	55.642
MSE 0.3 e20	88.307	77.401	63.396	72.099	63.316	55.827
MSE 0.3 e40	88.267	77.345	63.296	72.174	63.345	55.830
MSE 3 e20	88.494	77.456	63.513	72.376	63.429	56.055
MSE 3 e40	88.508	77.563	63.552	72.258	63.468	55.970
CWD 0.03 e20	88.369	77.296	63.318	71.988	63.113	55.701
CWD 0.03 e40	88.062	77.222	63.239	71.828	63.037	55.605
CWD 0.3 e20	88.248	77.386	63.460	72.406	63.423	55.966
CWD 0.3 e40	88.318	77.363	63.362	72.306	63.404	55.966
CWD 3 e20	88.206	77.384	63.508	73.009	63.785	56.210
CWD 3 e40	88.157	77.437	63.553	73.101	63.976	56.392

The primary native KITTI summary is the arithmetic mean of Car M and Ped M; all remaining difficulty scores are preserved here. Detection outputs came from the same immutable KITTI prediction archive used for the common views. The numerical audit checked exact export/label bindings and recomputed AP_R40 from saved precision curves; it did not rerun native C++ matching.

S5 All secondary epoch contrasts

Table S7. Epoch-20 paired differences, mean ± sample SD in percentage points. This is the same 40-epoch schedule, not a separate 20-epoch experiment. All primary epoch-40 contrasts are in manuscript Tables 1 and 2.

Objective coefficient	BDD native	BDD common	KITTI KI	KITTI KN	KITTI native
MSE 0.03	-0.000 ± 0.015	0.014 ± 0.039	0.002 ± 0.018	-0.007 ± 0.010	-0.006 ± 0.047
MSE 0.3	-0.032 ± 0.013	0.018 ± 0.036	0.016 ± 0.015	0.016 ± 0.040	0.057 ± 0.168
MSE 3	-0.277 ± 0.088	-0.113 ± 0.039	-0.003 ± 0.101	0.003 ± 0.093	0.035 ± 0.257
CWD 0.03	-0.008 ± 0.006	0.016 ± 0.037	0.003 ± 0.031	-0.003 ± 0.031	-0.004 ± 0.020
CWD 0.3	-0.008 ± 0.001	0.012 ± 0.031	0.045 ± 0.063	0.042 ± 0.071	0.078 ± 0.103
CWD 3	-0.213 ± 0.121	-0.058 ± 0.046	0.031 ± 0.111	0.024 ± 0.104	0.044 ± 0.287

Table S8. Native BDD paired differences for epoch 20 in each starting pair, percentage points. Three pairs remain the sample size.

Objective coefficient	Pair 101	Pair 202	Pair 303
MSE 0.03	+0.0163	-0.0101	-0.0075
MSE 0.3	-0.0447	-0.0193	-0.0334
MSE 3	-0.2098	-0.3772	-0.2452
CWD 0.03	-0.0009	-0.0130	-0.0090
CWD 0.3	-0.0084	-0.0076	-0.0087
CWD 3	-0.1660	-0.3507	-0.1224

For every other metric, the three raw paired differences are recoverable directly from Tables S1–S6 and are provided without rounding in all_distillation_vs_control_contrasts. Mean ± SD is descriptive across the three pairs. Neither a confidence interval nor a test of no effect is implied.

S6 Objective scale and training diagnostics

The first recorded batch supplies an exact weighted distillation-to-detection objective ratio. This is not a gradient ratio. Equal coefficients correspond to different loss scales. Saved first-batch snapshots were newly byte-bound during extraction; they were outside the earlier full saved-state audit and do not reconstruct historical live gradients.

Table S9. First-batch weighted KD/detection loss ratio, mean ± sample SD across pairs. Also shown are epoch-40 detection and feature loss means. Detection is a mean of batch sums; feature loss is a mean of sums of per-scale means.

Objective coefficient	First-batch ratio	Epoch 40 detection loss	Epoch 40 feature loss
control 0	0.00000 ± 0.00000	185.7177 ± 0.1072	0.0000 ± 0.0000
cwd 0.03	0.00721 ± 0.00123	185.7127 ± 0.1083	0.6678 ± 0.0946
cwd 0.3	0.07215 ± 0.01234	185.8880 ± 0.0783	0.4113 ± 0.0695
cwd 3	0.72146 ± 0.12341	186.8493 ± 0.1150	0.2442 ± 0.0050
mse 0.03	0.00967 ± 0.00039	185.7245 ± 0.1217	0.7358 ± 0.0183
mse 0.3	0.09674 ± 0.00391	185.9130 ± 0.1062	0.5277 ± 0.0088
mse 3	0.96735 ± 0.03905	187.1295 ± 0.1395	0.3550 ± 0.0050

The last batch contains 39 images, while the other batches contain 64. The two epoch averages alone do not determine an exact epoch weighted-KD/detection ratio or an average of batch ratios. We therefore do not infer either. Epochs 1 and 20, all 840 epoch records summarized per trajectory, raw pair values and normalization notes are retained in completed_training_diagnostics_summary.json.

The 21 trainer wall times sum to 284,735.77 seconds, including diagnostic validation and state writes while other jobs ran concurrently. This is neither elapsed calendar duration nor an isolated GPU compute or inference benchmark. No energy or speedup claim is made from it.

S7 Applied and skipped updates

Across all continuations: 917,280 attempts = 916,981 applied updates + 299 skipped updates. The saved EMA update count matches the applied count for each trajectory. Skipped AMP steps are retained, not counted as successful parameter updates. Base runs did not retain equivalent per-step counters.

S8 Verification scope and retained failures

All six base final raw states, EMA states and optimizer tensors were audited. Within each architecture, starting and final parameter-value digests differed across the three seeds. Nano runs had 183 changed raw parameter tensors and large runs had 303; fixed DFL projections stayed unchanged. All 21 continuations showed changed student parameters and finite final model, adapters, EMA and optimizer states. Final checkpoints exactly matched the saved EMA after conversion to their serialization precision.

The final full-precision sidecars allow this comparison at epoch 40. Intermediate epoch-20 raw EMA sidecars were not retained. Those checkpoints are finite, architecture-checked and bound to trainer hashes, but the full-precision intermediate EMA cannot be independently reconstructed. For fixed DFL EMA values, tiny float32 accumulation differences were explained by a replay of the native EMA rule; they were not optimizer updates to frozen student parameters.

There are 281 undefined or nonfinite epoch gradient-median records. They remain explicit in the diagnostic output. The underlying per-batch norms were not archived, so this does not identify a frequency of nonfinite gradients, and it does not justify a claim that every backward pass was finite. Recorded finite epoch losses and final state checks have a different scope.

Technical recovery record

Before full training, a module-name collision and the fixed native loader generator were detected and corrected; original failed preflights remain in technical_failures. A base cache construction race was recovered under the unchanged protocol without repeating completed bases. Monitoring assumptions about exact EMA DFL equality, gradient medians and normal coordinator exit were corrected in separate audit records, preserving the original checks. None constitutes a discarded unfavorable experimental outcome.

During evaluation, the initial GPU command selected physical GPU 0 despite a masking environment variable. The two already-started native BDD jobs finished and their outputs were retained byte-for-byte. A separate scheduler version corrected physical-device selection for unfinished work. The cached large-pair-101 native BDD result is the documented GPU assignment exception. Scientific inference settings did not change, completed inference was not repeated, and no timing comparison is based on the affected jobs.

The first aggregation attempt used a folder name where a proof-field name differed. It failed before writing a result. A separate corrected source version mapped kitti_native to native_kitti and passed all input-hash and completeness checks. The original source and failure log remain in audit_technical_failures/004_aggregation_artifact_key. A second independent arithmetic implementation reproduced all final summaries. No scores, checkpoints or method settings were altered to resolve this reporting error.

Audit limits and reproducibility

Common scores were replayed with the same pinned COCO implementation. Native BDD numerical replay uses stored true-positive statistics instead of independently reconstructing IoU matches from images. Native KITTI replay recomputes integration from saved precision curves; it does not independently reproduce C++ matching. Source bindings and synthetic tests add evidence but do not create an independent implementation. Full raw datasets and binary prediction/model archives are not included in this public text/JSON package. Author review must settle durable access before submission.